Tertiary Infotech Academy Pte Ltd

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LESSON PLAN

For

Certified Kubernetes Application Developer (CKAD)

TGS Ref No: TGS-2025053212

Conducted by

Tertiary Infotech Academy Pte Ltd

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Version 5.0

DOCUMENT VERSION CONTROL RECORD

Version Number	Effective Date of Release	Summary of Included Changes	Author
1.0	30 June 2026	First version — 3½-day lesson plan aligned to the CKAD slide decks and all 30 hands-on labs; Days 1–3: 9:00 AM – 6:00 PM; Day 4: half-day + assessment from 2:00 PM	Tertiary Infotech Academy Pte Ltd
5.0	28 Jul 2026	Realigned to CKAD slide deck v5.0 (555 slides); learning outcomes aligned to the registered competency standard.	Tertiary Infotech Academy

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Course Title	Certified Kubernetes Application Developer (CKAD)
Course Code	TGS-2025053212
Duration	3½ Days (28 training hours)
Daily Schedule (Days 1–3)	9:00 AM – 6:00 PM (8 training hours/day, excluding lunch)
Day 4 Schedule	9:00 AM – 12:00 PM training, then lunch, mock-exam practice & mock assessment 1:00 –3:30 PM, tea break, final assessment briefing & assessment 3:45–6:00 PM
Lunch Break	1:00 PM – 2:00 PM (1 hour, Days 1–3) · 12:00 PM – 1:00 PM (Day 4)
Breaks	Short tea breaks scheduled within each day's training hours
Delivery Mode	Instructor-led, hands-on labs in Killercoda browser terminals
Prerequisites	Basic Linux command-line familiarity; exposure to containers helpful but not required
Tools	kubectl, Kubernetes cluster (Killercoda / minikube / kind), Docker, Helm 3, killer.sh

Course Overview

This 3½-day hands-on course prepares participants for the Certified Kubernetes Application Developer (CKAD) certification exam. Working entirely in Killercoda browser terminals, learners build real Kubernetes workloads across all five CKAD exam domains: Application Design & Build (Day 1), Application Deployment (Day 2), Observability & Maintenance and Configuration & Security (Day 3), and Services & Networking (Day 4 morning). The course emphasises exam-speed kubectl techniques — `--dry-run=client -o yaml` manifests, `alias k=kubectl`, `kubectl explain` and `kubectl auth can-i` — alongside conceptual understanding aligned to the official kubernetes.io documentation.

Learning Outcomes

By the end of this course, participants will be able to:

- Build and optimise container images with single- and multi-stage Dockerfiles.
- Create and manage Pods imperatively and declaratively using kubectl with --dry-run scaffolding.
- Run batch and scheduled workloads with Jobs and CronJobs.
- Design multi-container Pods using the sidecar and init-container patterns.
- Persist and share Pod data with emptyDir, tmpfs and hostPath volumes.
- Deploy, scale and roll back applications using Deployments; apply blue/green and canary strategies.
- Package and deploy applications with Helm 3 charts and Kustomize overlays.
- Manage cluster-wide workloads with DaemonSets and StatefulSets.
- Configure liveness, readiness and startup probes and diagnose failing workloads.

- Collect and interpret logs and metrics using kubectl logs, kubectl top and the Metrics API.
- Inject non-secret and sensitive configuration using ConfigMaps and Secrets.
- Harden containers with securityContext, ServiceAccounts and RBAC Role/RoleBinding.
- Govern namespace resource consumption with ResourceQuota and LimitRange.
- Expose applications internally and externally using ClusterIP, NodePort and Ingress.
- Isolate Pod-to-Pod traffic with NetworkPolicy allow-list rules.

Day 1

Domain 1 — Application Design and Build (CKAD 20%)

Morning Session · 9:00 AM – 1:00 PM

Time	Topic / Activity	Duration	Slides
9:00 – 9:15	Welcome, course objectives & overview (TGS-2025053212)	15 min	2–12
9:15 – 9:40	**Topic 1: Container Images — Dockerfiles, layers & multi-stage builds**	25 min	13–18
9:40 – 10:10	Lab 1: Build a Container Image with Docker	30 min	19–22
10:10 – 10:40	Lab 2: Multi-Stage Dockerfile	30 min	23–26
10:40 – 10:55	***Tea Break***	15 min	—
10:55 – 11:15	**Topic 2: Pods — the exam-speed kubectl workflow (`alias k`, `$do`)**	20 min	28–33
11:15 – 11:45	Lab 3: Create and Manage Pods	30 min	34–37
11:45 – 12:15	Lab 4: Jobs — Run-to-Completion Workloads	30 min	39–48
12:15 – 12:45	Lab 5: CronJobs — Scheduled Workloads	30 min	49–53
12:45 – 1:00	Morning Q&A	15 min	—
Time	**Topic / Activity**	**Duration**	Slides
1:00 – 2:00	***Lunch Break***	60 min	38

Afternoon Session · 2:00 PM – 6:00 PM

Time	Topic / Activity	Duration	Slides
2:00 – 2:20	**Topic 3: Multi-Container Pods — sidecar & init-container patterns**	20 min	55–62
2:20 – 2:55	Lab 6: Multi-Container Pods — Sidecar Pattern	35 min	63–66
2:55 – 3:30	Lab 7: Init Containers	35 min	67–70
3:30 – 3:45	***Tea Break***	15 min	—
3:45 – 4:05	**Topic 4: Volumes — emptyDir, tmpfs & hostPath**	20 min	71–75
4:05 – 4:45	Lab 8: Volumes — emptyDir and hostPath	40 min	76–79
4:45 – 5:25	kubectl dry-run drills & manifest-speed practice	40 min	—
5:25 – 5:55	Open lab / Q&A	30 min	—
5:55 – 6:00	Day 1 recap & wrap-up	5 min	80

Day 2

Domain 2 — Application Deployment (CKAD 20%)

Morning Session · 9:00 AM – 1:00 PM

Time	Topic / Activity	Duration	Slides
9:00 – 9:15	Day 1 recap & Day 2 objectives	15 min	82–84
9:15 – 9:40	**Topic 5: Deployments & ReplicaSets — scale, self-heal, rollout**	25 min	102–105
9:40 – 10:15	Lab 9: Deployments and ReplicaSets	35 min	106–109
10:15 – 10:50	Lab 10: Rolling Updates and Rollback	35 min	111–118
10:50 – 11:05	***Tea Break***	15 min	—
11:05 – 11:25	**Topic 6: Release Strategies — blue/green & canary**	20 min	120–123
11:25 – 12:00	Lab 11: Blue/Green Deployment	35 min	124–127
12:00 – 12:35	Lab 12: Canary Deployment	35 min	128–131
12:35 – 1:00	Morning Q&A	25 min	—
Time	**Topic / Activity**	**Duration**	**Slides**
1:00 – 2:00	***Lunch Break***	60 min	119

Afternoon Session · 2:00 PM – 6:00 PM

Time	Topic / Activity	Duration	Slides
2:00 – 2:20	**Topic 7: Helm — Kubernetes package manager**	20 min	139–142
2:20 – 3:00	Lab 13: Helm — Install, Upgrade, Rollback	40 min	143–146
3:00 – 3:20	**Topic 8: Kustomize — overlay-based configuration**	20 min	147–150
3:20 – 4:00	Lab 14: Kustomize Overlays	40 min	151–155
4:00 – 4:15	***Tea Break***	15 min	—
4:15 – 4:35	**Topic 9: DaemonSets & StatefulSets**	20 min	157–160
4:35 – 5:20	Lab 15: DaemonSets and StatefulSets	45 min	161–164
5:20 – 5:55	Review & Domain 2 consolidation	35 min	—
5:55 – 6:00	Day 2 recap & wrap-up	5 min	165

Day 3

Domains 3 & 4 — Observability & Maintenance + Configuration & Security (CKAD 40%)

Morning Session · 9:00 AM – 1:00 PM (Domain 3 — Observability & Maintenance)

Time	Topic / Activity	Duration	Slides
9:00 – 9:15	Day 2 recap & Day 3 objectives	15 min	167–169
9:15 – 9:35	**Topic 10: Health Probes — liveness, readiness & startup**	20 min	171–175
9:35 – 10:10	Lab 16: Liveness, Readiness & Startup Probes	35 min	176–179
10:10 – 10:40	Lab 17: Container Logging	30 min	180–186
10:40 – 10:55	***Tea Break***	15 min	—
10:55 – 11:15	**Topic 11: Metrics & Debugging — kubectl top, events & ephemeral containers**	20 min	188–197
11:15 – 11:45	Lab 18: kubectl top and Metrics Server	30 min	191–194
11:45 – 12:20	Lab 19: Debugging Pods and Events	35 min	198–201
12:20 – 12:50	Lab 20: API Deprecations & kubectl explain	30 min	202–208
12:50 – 1:00	Morning Q&A	10 min	—
Time	**Topic / Activity**	**Duration**	**Slides**
1:00 – 2:00	***Lunch Break***	60 min	209

Afternoon Session · 2:00 PM – 6:00 PM (Domain 4 — Configuration & Security)

Time	Topic / Activity	Duration	Slides
2:00 – 2:20	**Topic 12: ConfigMaps & Secrets — inject config & sensitive data**	20 min	210–214
2:20 – 2:55	Lab 21: ConfigMaps — Env & Volume Injection	35 min	215–218
2:55 – 3:30	Lab 22: Secrets	35 min	219–226
3:30 – 3:45	***Tea Break***	15 min	—
3:45 – 4:05	**Topic 13: SecurityContext & ServiceAccounts — identity & least privilege**	20 min	228–230
4:05 – 4:40	Lab 23: SecurityContext	35 min	231–234
4:40 – 5:05	Lab 24: ServiceAccounts	25 min	235–241
5:05 – 5:35	Lab 25: RBAC — Roles & RoleBindings	30 min	242–249
5:35 – 5:55	Lab 26: ResourceQuota & LimitRange	20 min	267–274
5:55 – 6:00	Day 3 recap & wrap-up	5 min	287

Day 4 (Half Day)

Domain 5 — Services & Networking (CKAD 20%) + Final Assessment from 3:45 PM

Morning Training · 9:00 AM – 12:00 PM

Time	Topic / Activity	Duration	Slides
9:00 – 9:10	Day 3 recap & Day 4 objectives	10 min	289–291
9:10 – 9:30	**Topic 14: Services — ClusterIP, NodePort, LoadBalancer & Service DNS**	20 min	292–298
9:30 – 10:05	Lab 27: Services — ClusterIP, NodePort, LoadBalancer	35 min	299–302
10:05 – 10:35	Lab 28: Service DNS	30 min	311–317
10:35 – 10:50	***Tea Break***	15 min	—
10:50 – 11:10	**Topic 15: Ingress & TLS + NetworkPolicy — L7 routing & Pod isolation**	20 min	319–330
11:10 – 11:40	Lab 29: Ingress with TLS	30 min	322–326
11:40 – 12:00	Lab 30: NetworkPolicy	20 min	331–335
Time	**Topic / Activity**	**Duration**	**Slides**
12:00 – 1:00	***Lunch Break***	60 min	337

Afternoon · 1:00 PM – 6:00 PM

Time	Topic / Activity	Duration	Slides
1:00 – 3:30	**Mock-Exam Practice & Mock Assessment — killer.sh / Killercoda scenarios across all 5 domains**	150 min	336–340
3:30 – 3:45	***Tea Break***	15 min	—
3:45 – 6:00	**Assessment Briefing & Final Assessment — written / MCQ on the LMS (open book; covers all 5 domains)**	135 min	341–344

Labs Covered

All 30 hands-on labs are delivered in Killercoda browser terminals and mirror the slide decks and Learner Guide exactly:

Lab	Domain	Topic	Title
1	Application Design and	Container images	Build a Container Image

	Build		with Docker
2	Application Design and Build	Container images	Multi-Stage Dockerfile
3	Application Design and Build	Pods	Create and Manage Pods
4	Application Design and Build	Batch workloads	Jobs — Run-to-Completion Workloads
5	Application Design and Build	Batch workloads	CronJobs — Scheduled Workloads
6	Application Design and Build	Multi-container Pods	Multi-Container Pods — Sidecar Pattern
7	Application Design and Build	Multi-container Pods	Init Containers
8	Application Design and Build	Volumes	Volumes — emptyDir and hostPath
9	Application Deployment	Deployments & ReplicaSets	Deployments and ReplicaSets
10	Application Deployment	Rolling Updates & Rollback	Rolling Updates and Rollback
11	Application Deployment	Blue/Green Deployment	Blue/Green Deployment
12	Application Deployment	Canary Deployment	Canary Deployment
13	Application Deployment	Helm	Helm — Install, Upgrade, Rollback
14	Application Deployment	Kustomize	Kustomize Overlays
15	Application Deployment	DaemonSets & StatefulSets	DaemonSets and StatefulSets
16	Observability & Maintenance	Probes	Liveness, Readiness & Startup Probes
17	Observability & Maintenance	Logging	Container Logging
18	Observability & Maintenance	Metrics	kubectl top and Metrics Server
19	Observability & Maintenance	Debugging	Debugging Pods and Events
20	Observability & Maintenance	API Deprecations	API Deprecations & kubectl explain
21	Configuration & Security	ConfigMaps	ConfigMaps — Env & Volume Injection
22	Configuration & Security	Secrets	Secrets
23	Configuration & Security	SecurityContext	SecurityContext
24	Configuration & Security	ServiceAccounts	ServiceAccounts
25	Configuration & Security	RBAC	RBAC — Roles & RoleBindings
26	Configuration & Security	Quotas & Limits	ResourceQuota & LimitRange
27	Services & Networking	Services	Services — ClusterIP, NodePort, LoadBalancer
28	Services & Networking	Service DNS	Service DNS
29	Services & Networking	Ingress & TLS	Ingress with TLS
30	Services & Networking	NetworkPolicy	NetworkPolicy

Assessment

Participants are assessed through their hands-on lab work across Days 1–4 and a final written / MCQ assessment on Day 4 afternoon (from 3:45 PM) delivered on the LMS. The assessment is open book — slides, the Learner Guide and kubernetes.io are permitted. A minimum of 75% attendance across all sessions is required for SSG funding eligibility.